# RECEIPT SCANNER

VOLUME 1

## SOFTWARE REQUIREMENTS

# RECEIPT SCANNER

## 28 MARCH 2026

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**Project:** Receipt Scanner V2

**Document Class:** SRS

**Version:** 0.1.2

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# RECEIPT SCANNER

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This work has been completed under the direction of Dr. Natalia Bell, Marymount University, as part of the coursework for IT-548-A, and is submitted in fulfillment of academic requirements for the specified class.

James Green, 2024.

# TABLE OF CONTENTS

**RECEIPT SCANNER** ..... **I**

**TABLE OF CONTENTS** ..... **II**

**LIST OF TABLES/FIGURES** ..... **IV**

**CHAPTER 1 INTRODUCTION** ..... **1-1**

1.1 Background ..... 1-1
1.2 Scope ..... 1-2
1.3 Document Overview ..... 1-3

**CHAPTER 2 SYSTEM OVERVIEW** ..... **2-1**

2.1 Product Perspective ..... 2-1
2.2 Product Functions ..... 2-1
2.3 User Classes and Characteristics ..... 2-2
2.4 Operating Environment ..... 2-2
2.5 Design and Implementation Constraints ..... 2-3
2.6 Assumptions and Dependencies ..... 2-3

**CHAPTER 3 FUNCTIONAL REQUIREMENTS** ..... **3-1**

**CHAPTER 4 EXTERNAL INTERFACE REQUIREMENTS** ..... **4-1**

4.1 User Interfaces ..... 4-1
4.2 Hardware Interfaces ..... 4-1
4.3 Software Interfaces ..... 4-2
4.4 Communication Interfaces ..... 4-2

**CHAPTER 5 NON-FUNCTIONAL REQUIREMENTS** ..... **5-1**

**CHAPTER 6 DATA REQUIREMENTS** ..... **6-1**

6.1 Overview ..... 6-1
6.2 Receipt Data Structure ..... 6-1
6.3 Receipt Collection Data Structure ..... 6-2
6.4 Collection Summary Data Structure ..... 6-2
6.5 Data Relationships ..... 6-3
6.6 Data Persistence ..... 6-3
6.7 Data Constraints ..... 6-3

**CHAPTER 7 VERIFICATION / VALIDATION CRITERIA** ..... **7-1**

7.1 Overview ..... 7-1
7.2 Verification Methods ..... 7-1
7.3 Verification Table ..... 7-2

**CHAPTER 8 RTM** ..... **8-1**

8.1 Overview ..... 8-1
8.2 RTM ..... 8-2

**APPENDIX A** ..... **A-1**

# LIST OF TABLES/FIGURES

**TABLE 3-1 FUNCTIONAL REQUIREMENTS** ..... 3-1
**TABLE 5-1 NON-FUNCTIONAL REQUIREMENTS** ..... 5-1
**TABLE 7-1 FUNCTIONAL REQUIREMENT VERIFICATION TABLE** ..... 7-2
**TABLE 8-1 REQUIREMENTS TRACEABILITY MATRIX** ..... 8-2

# CHAPTER 1
# INTRODUCTION

## 1.1 Background

The increasing reliance on digital tools for personal and small-scale financial management has created a need for efficient methods of capturing, organizing, and analyzing transaction data. Receipts, which serve as primary records of financial activity, are often retained in physical form, making them difficult to aggregate, search, and analyze over time. Manual entry of receipt data into digital systems is time-consuming, error-prone, and inefficient, particularly for users who manage recurring expenses such as fuel, travel, or operational costs.

The original receipt scanner software focused on the processing of individual receipts of a single class: gas receipts, and was developed as an academic group project to explore the use of Optical Character Recognition (OCR) for extracting structured data from receipt images. That system demonstrated the feasibility of automating receipt data extraction but was limited in scope, capability, and scalability.

This document is part of a collection of five volumes, which collectively describe the second iteration of the Receipt Scanner system, developed as an individual effort. This version expands upon the original implementation by generalizing receipt processing capabilities, improving usability, and introducing additional features to support broader financial tracking and analysis use cases. Volume 1 describes the requirements of the software. Volume 2 describes the design of the system. Volume 3 consists of the codebase for this software. Volume 4 describes the testing of the software, and in conjunction with Volume 1, establishes a traceability matrix. Volume 5 consists of a user guide describing the usage of the software to the end user.

## 1.2 Scope

The Receipt Scanner (Version 2) system is designed to provide users with a streamlined method for digitizing, processing, and analyzing receipt data. The system enables users to upload receipt images, extract relevant financial information using OCR-based processing, and organize that information into structured formats for review and export.

This iteration extends the capabilities of the original system in several key areas. The system supports the upload and processing of multiple receipts within a single workflow, allowing for improved efficiency in batch operations. It generalizes receipt handling beyond gas receipts to accommodate a wider range of receipt types. To support this, the system introduces the ability for users to specify receipt categories, enabling the selection of appropriate OCR processing logic tailored to different receipt formats.

Additionally, the system provides enhanced data analysis capabilities, including the generation of month-to-month summaries of financial activity. Users are also able to export processed receipt data in a structured format, such as CSV, to support external analysis or record-keeping.

The system is intended for individual users or small-scale use cases and is implemented as a lightweight application with a focus on usability and extensibility. It does not aim to replace enterprise financial systems but rather to provide a practical tool for receipt digitization and basic financial tracking.

## 1.3 Document Overview

This document presents the Software Requirements Specification (SRS) for the Receipt Scanner (Version 2) system. It is organized to provide a structured description of the system's requirements in accordance with established software engineering standards.

Chapter 2 provides an overall system overview, including the system's context, major functions, user characteristics, and operational constraints. Chapter 3 defines the functional requirements of the system, detailing the specific behaviors and capabilities the system must support. Chapter 4 describes external interface requirements, including user interfaces and system interactions.

Chapter 5 outlines non-functional requirements, such as performance, usability, and reliability considerations. Chapter 6 defines data requirements, including data structures and relationships. Chapter 7 specifies verification and validation criteria to ensure that system requirements can be tested and confirmed. Finally, Chapter 8 presents the Requirements Traceability Matrix (RTM), which maps requirements to their corresponding implementation and validation artifacts.

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# CHAPTER 2
# SYSTEM OVERVIEW

## 2.1 Product Perspective

The Receipt Scanner (Version 2) system is a standalone application designed to digitize process, and analyze receipt data for individual users. It represents an evolution of the original Receipt Scanner academic group project, expanding its capabilities while maintaining its core objective of automating receipt data extraction.

From a system perspective, the application functions as a data processing and management tool that integrates user input (receipt images) with Optical Character Recognition (OCR) technologies to produce structured financial data. The system operates as a self-contained solution, with minimal external dependencies beyond OCR processing libraries and standard runtime environments.

This version introduces a more generalized processing model, enabling support for multiple receipt types (such as parking, food, airfare, etc), and batch operations. The system architecture is designed to be modular, allowing for the selection of different processing logic based on user-specified receipt categories. This supports extensibility for future enhancements, such as additional receipt types or improved extraction models.

## 2.2 Product Functions

At a high level, the system provides the following core functions:

- Upload and manage one or more receipt images within a single session
- Process receipt images using OCR to extract relevant financial data
- Support multiple receipt types through user selection, enabling tailored extraction logic
- Store and organize extracted data into structured formats
- Generate summarized financial data on a month-to-month basis
- Export processed receipt data into a CSV format for external use

These functions collectively enable the transformation of unstructured receipt data into usable financial information, supporting both record-keeping and basic analysis.

## 2.3 User Classes and Characteristics

The system is intended for individual users with basic technical proficiency. Primary user classes include:

- **General Users**: Individuals seeking to digitize and organize personal or small-business receipts. These users are expected to have minimal technical expertise and require an intuitive interface.
- **Advanced Users**: Users with familiarity in data handling or financial tracking who may utilize exported data for further analysis in external tools such as spreadsheets or financial software.

Users are not required to have prior knowledge of OCR technologies or system internals. The system is designed to minimize user interaction complexity while providing clear feedback during processing operations.

## 2.4 Operating Environment

The system is designed to operate within a standard computing environment with the following characteristics:

- A system capable of running a Python-based application
- A local or web-based interface accessible through a modern web browser
- Sufficient system resources to perform OCR processing on uploaded images
- File system access for storing uploaded receipt images and processed data

The system may optionally be deployed as a lightweight local web server, enabling browser-based interaction without requiring complex installation or configuration.

## 2.5 Design and Implementation Constraints

The system is subject to the following constraints:

- **Technology Constraints**: The implementation is primarily Python-based, limiting design choices to compatible libraries and frameworks. OCR capabilities are dependent on the accuracy and limitations of the selected OCR engine.
- **Time Constraints**: As an academic project, development is constrained by a limited timeline, which may impact the scope of advanced features or optimizations.
- **Resource Constraints**: The system is intended to run in a lightweight environment, restricting the use of computationally intensive models or large-scale infrastructure.
- **Data Quality Constraints**: The accuracy of extracted data is dependent on the quality and variability of receipt images, including factors such as resolution, formatting, and noise.

## 2.6 Assumptions and Dependencies

The system is developed under the following assumptions:

- Users will provide receipt images of sufficient quality for OCR processing
- Receipt formats, while variable, will contain recognizable patterns that can be processed by the system
- Users will correctly identify or select the appropriate receipt type when required
- The system will be used in a non-adversarial context, without malicious input designed to disrupt processing

Key dependencies include:

- OCR libraries or engines used for text extraction
- Python runtime environment and associated dependencies
- Local or browser-based interface components for user interaction

These assumptions and dependencies define the operational boundaries within which the system is expected to function effectively.

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# CHAPTER 3

# FUNCTIONAL REQUIREMENTS

**Table 3-1 Functional Requirements**

| ID | Requirement Description | Input | Processing | Output |
|---|---|---|---|---|
| **FR-1** | The system shall allow users to upload one or more receipt images in a single session. | Receipt Image File(s) | Accept and Register uploaded Files | Receipts available in system |
| **FR-2** | The system shall validate uploaded files to ensure they are supported image formats. | Receipt image file(s) | Check file type and format | Validated or rejected file(s) |
| **FR-3** | The system shall allow users to select a receipt type prior to processing. | User selection | Associate receipt type with input | Receipt categorized |
| **FR-4** | The system shall process receipt images using OCR to extract text data. | Receipt image | Apply OCR engine | Extracted text |
| **FR-5** | The system shall apply receipt-type-specific parsing logic to extracted text. | OCR text + receipt type | Identify relevant fields | Parsed receipt data |
| **FR-6** | The system shall extract and structure key receipt data, including date, total amount, and vendor. | Parsed data | Map to structured format | Structured receipt record |
| **FR-7** | The system shall support batch processing of multiple receipts within a single workflow. | Multiple receipt images | Iterate processing pipeline | Multiple processed records |
| **FR-8** | The system shall display processed receipt data to the user. | Structured receipt data | Format for display | Visible receipt details |
| **FR-9** | The system shall allow users to edit extracted receipt data. | Structured receipt data | Apply user modifications | Updated receipt data |
| **FR-10** | The system shall store processed receipt data for later access. | Structured receipt data | Persist data | Stored dataset |
| **FR-11** | The system shall generate monthly summaries of receipt data. | Stored receipt data | Group and aggregate by month | Monthly summary |
| **FR-12** | The system shall calculate aggregate financial values, including totals and averages where applicable. | Receipt dataset | Perform calculations | Computed metrics |

**Table 3-1 Functional Requirements**

| ID | Requirement Description | Input | Processing | Output |
|---|---|---|---|---|
| **FR-13** | The system shall allow users to export receipt data in CSV format. | Stored receipt data | Convert to CSV | Downloadable CSV file |
| **FR-14** | The system shall allow users to exclude selected receipts from summary calculations. | Receipt selection | Filter dataset | Adjusted summary |
| **FR-15** | The system shall provide feedback on processing status to the user. | Processing events | Track status | Status messages |
| **FR-16** | The system shall handle OCR and parsing errors and notify the user of failures. | OCR/processing results | Detect and handle errors | Error notification |

# CHAPTER 4

# EXTERNAL INTERFACE REQUIREMENTS

## 4.1 User Interfaces

The system shall provide a user interface that enables users to interact with receipt data in a clear and efficient manner. The interface shall be accessible through a web-based or locally hosted application and shall support standard user interactions.

The user interface shall include functionality to:

- Upload one or more receipt images
- Select receipt type prior to processing
- View processed receipt data in a structured format
- Edit extracted receipt data as needed
- View aggregated summaries of receipt data
- Export receipt data in CSV format

The interface shall present feedback to the user regarding processing status, including success and failure conditions. The design shall prioritize usability and clarity to support users with minimal technical expertise.

## 4.2 Hardware Interfaces

The system shall interface with standard user hardware, including:

- Local storage devices for receipt image upload and data storage
- Input devices such as keyboard and mouse for user interaction

No specialized hardware interfaces are required. The system shall operate on general-purpose computing systems capable of running the application and associated dependencies.

## 4.3 Software Interfaces

The system shall interface with the following software components:

- OCR processing libraries or engines used for text extraction from receipt images
- The Python runtime environment required to execute the application
- Web browser software used to access the user interface (if deployed as a web application)

The system shall ensure compatibility with supported libraries and runtime environments necessary for OCR processing and application execution

## 4.4 Communication Interfaces

The system shall support communication between the user interface and the processing components of the application. If deployed as a web-based system, this communication shall occur over standard HTTPS protocols.

No external network communication with third-party services is required for core functionality. All processing may be performed locally unless otherwise extended in future iterations.

# CHAPTER 5

# NON-FUNCTIONAL REQUIREMENTS

**Table 5-1 Non-Functional Requirements**

| ID | Description | Category |
|---|---|---|
| **NFR-1** | The system shall process individual receipt images within a reasonable time under normal operating conditions. | Performance |
| **NFR-2** | The system shall support batch processing of multiple receipts without system failure. | Performance |
| **NFR-3** | The system shall provide user feedback within a reasonable time after receipt upload and processing actions. | Performance |
| **NFR-4** | The system shall provide an intuitive and easy-to-use interface for users with minimal technical expertise. | Usability |
| **NFR-5** | The system shall display receipt data in a clear and readable format. | Usability |
| **NFR-6** | The system shall allow users to easily correct or modify extracted data. | Usability |
| **NFR-7** | The system shall handle invalid inputs and processing errors without crashing. | Reliability |
| **NFR-8** | The system shall maintain consistent operation during standard usage conditions. | Reliability |
| **NFR-9** | The system shall ensure that stored receipt data remains accessible during system operation. | Reliability |
| **NFR-10** | The system shall restrict processing to user-provided inputs and not execute unauthorized operations. | Security |
| **NFR-11** | The system shall not expose user data outside the system without user-initiated actions (e.g., export). | Security |

**Table 5-1 Non-Functional Requirements**

| ID | Description | Category |
|---|---|---|
| **NFR-12** | The system shall be compatible with standard computing environments capable of running a Python-based application. | Compatibility |
| **NFR-13** | The system shall operate within the constraints of available system memory and processing resources. | Performance |
| **NFR-14** | The system shall allow future extension to support additional receipt types and processing logic. | Maintainability |
| **NFR-15** | The system shall be structured to allow modification of OCR and parsing components without requiring full system redesign. | Maintainability |

# CHAPTER 6
# DATA REQUIREMENTS

## 6.1 Overview

The *Receipt Scanner (Version 2)* system processes and manages data derived from user-provided receipt images. The system transforms unstructured image data into structured financial records, which are stored, displayed, and optionally exported.

The primary data entities within the system include individual receipt records, collections of receipts, and aggregated summary data. These entities support core system functionality such as receipt processing, user review, financial summarization, and data export.

## 6.2 Receipt Data Structure

A receipt represents a single transaction extracted from an uploaded receipt image. Each receipt record shall contain structured fields derived from OCR and parsing processes.

At a minimum, each receipt shall include:

- Vendor name
- Transaction date
- Total amount
- Receipt type (e.g., gas, retail, other)
- Raw OCR text (optional, for traceability and error handling)
- Processing status (e.g., valid, needs review, error)

Additional fields may be included depending on receipt type and parsing capabilities.

## 6.3 Receipt Collection Data Structure

The system shall maintain a collection of receipt records representing all processed receipts within the system's scope. This collection enables batch operations, aggregation, and data management.

The receipt collection shall:

- Store multiple receipt records
- Support addition of new receipts through upload and processing
- Allow modification of existing receipt records
- Allow exclusion of selected receipts from aggregation operations
- Maintain organization suitable for grouping and filtering

## 6.4 Collection Summary Data Structure

The system shall generate summary data derived from the receipt collection. This data supports financial analysis and reporting functions.

The summary data shall:

- Aggregate receipt data on a monthly basis
- Calculate total expenditures per time period
- Support calculation of derived metrics such as averages where applicable
- Reflect inclusion or exclusion of selected receipt records

The summary structure shall be suitable for both on-screen display and export.

## 6.5 Data Relationships

The system shall generate summary data derived from the receipt collection. This data supports financial analysis and reporting functions.

The summary data shall:

- Aggregate receipt data on a monthly basis
- Calculate total expenditures per time period
- Support calculation of derived metrics such as averages where applicable
- Reflect inclusion or exclusion of selected receipt records

The summary structure shall be suitable for both on-screen display and export.

## 6.6 Data Persistence

The system shall store receipt data in a manner that allows retrieval during system operation. Persistence mechanisms may include in-memory storage or local file-based storage, depending on system configuration.

- Stored data shall remain accessible for:
- User review and modification
- Aggregation and summary generation
- Export operations

## 6.7 Data Constraints

The system operates under the following data constraints:

- Receipt data accuracy is dependent on OCR quality and input image clarity
- Receipt formats may vary significantly, requiring flexible parsing logic
- Missing or incomplete data fields may occur and must be handled appropriately
- Numeric fields such as total amount shall be represented in a consistent format

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# CHAPTER 7

# VERIFICATION / VALIDATION CRITERIA

## 7.1 Overview

This section defines the methods used to verify and validate that the *Receipt Scanner (Version 2)* system satisfies the requirements specified in Chapter 3. Each functional requirement is associated with a verification approach to ensure that the system behaves as expected under defined conditions.

Verification methods include testing, inspection, and demonstration. The primary method used in this system is functional testing, where system behavior is evaluated against expected outputs.

## 7.2 Verification Methods

The following verification methods are used:

- **Test (T)**: Execution of the system with defined inputs to validate expected outputs
- **Inspection (I)**: Review of system outputs or artifacts without execution
- **Demonstration (D)**: Observation of system functionality in a controlled scenario

## 7.3 Verification Table

**Table 7-1 Functional Requirement Verification Table**

| Requirement ID | Verification Method | Test Description | Expected Result |
|---|---|---|---|
| FR-1 | T | Upload one or more valid receipt images | System accepts and stores receipt(s) |
| FR-2 | T | Upload unsupported file type (e.g., non-image) | System rejects file and provides feedback |
| FR-3 | T | Select receipt type prior to processing | System assigns selected type to receipt |
| FR-4 | T | Process receipt image through OCR | System produces extracted text |
| FR-5 | T | Process receipt with selected type | System applies appropriate parsing logic |
| FR-6 | T | Process valid receipt | System extracts date, total, and vendor fields |
| FR-7 | T | Upload and process multiple receipts | System processes all receipts without failure |
| FR-8 | D | View processed receipt data | System displays structured receipt information |
| FR-9 | T | Modify extracted receipt data | System updates stored receipt data |
| FR-10 | T | Process receipt and retrieve later | System retains and retrieves stored data |
| FR-11 | T | Generate monthly summary | System groups data and calculates totals |
| FR-12 | T | Generate summary with multiple receipts | System calculates aggregate values correctly |
| FR-13 | T | Export receipt data | System generates valid CSV file |
| FR-14 | T | Exclude receipt from summary | System omits selected receipt from calculations |
| FR-15 | D | Perform receipt processing | System provides status feedback to user |
| FR-16 | T | Process invalid or unreadable receipt | System detects error and notifies user |

# CHAPTER 8

# RTM

## 8.1 Overview

The Requirements Traceability Matrix (RTM) provides a mapping between system requirements, their corresponding specification sections, and verification methods. This ensures that all requirements are accounted for, implemented, and verifiable within the system.

The RTM supports traceability across the development lifecycle by linking each requirement to its validation criteria defined in Chapter 7.

## 8.2 RTM

**Table 8-1 Requirements Traceability Matrix**

| Requirement ID | Description | Source Section | Verification Reference | Test Case |
|---|---|---|---|---|
| FR-1 | Upload one or more receipt images | 3.1 | 7.3 | TC-1 |
| FR-2 | Validate supported file formats | 3.1 | 7.3 | TC-2 |
| FR-3 | Select receipt type prior to processing | 3.1 | 7.3 | TC-3 |
| FR-4 | Perform OCR text extraction | 3.1 | 7.3 | TC-4 |
| FR-5 | Apply receipt-type-specific parsing | 3.1 | 7.3 | TC-5 |
| FR-6 | Extract structured receipt data fields | 3.1 | 7.3 | TC-6 |
| FR-7 | Support batch processing of receipts | 3.1 | 7.3 | TC-7 |
| FR-8 | Display processed receipt data | 3.1 | 7.3 | TC-8 |
| FR-9 | Allow user modification of receipt data | 3.1 | 7.3 | TC-9 |
| FR-10 | Store and retrieve receipt data | 3.1 | 7.3 | TC-10 |
| FR-11 | Generate monthly summaries | 3.1 | 7.3 | TC-11 |
| FR-12 | Calculate aggregate financial values | 3.1 | 7.3 | TC-12 |
| FR-13 | Export receipt data to CSV | 3.1 | 7.3 | TC-13 |
| FR-14 | Exclude receipts from summaries | 3.1 | 7.3 | TC-14 |
| FR-15 | Provide processing status feedback | 3.1 | 7.3 | TC-15 |
| FR-16 | Handle OCR and parsing errors | 3.1 | 7.3 | TC-16 |

# APPENDIX A

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# NOMENCLATURE

| | |
|---|---|
| CSV | Comma-Separated Values |
| FR | Functional Requirement |
| Monthly Summary | Aggregated financial data grouped by month |
| NFR | Non-Functional Requirement |
| OCR | Optical Character Recognition |
| Receipt | A structured record representing a single transaction |
| Receipt Scanner | The system described in this document |
| RTM | Requirements Tracability Matrix |
| SRS | Software Requirement Specification |
| TC | Test Case |
| UI | User Interface |

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